Chapter 2

Evolution of the Pharmacy Profession and Public Health

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Learning Objectives:

Objective 2.1 Outline historical developments in the pharmacy profession.
Objective 2.2 Overview eras of pharmaceutical industries and their jobs for pharmacy graduates.
Objective 2.3 Discuss policies and laws governing the pharmacy profession and public health.
Objective 2.4 Detail an overview of public health and disease prevention.

OBJECTIVE 2.1. OUTLINE HISTORICAL DEVELOPMENTS IN THE PHARMACY PROFESSION

Pharmacists are healthcare professionals whose responsibilities and accountabilities include ensuring that people derive maximum therapeutic benefit and minimum harm from their treatments with drugs. They also work in disease prevention with or without medical interventions. It requires pharmacists to keep abreast of developments in:

- Clinical practice and the pharmaceutical sciences.
- Professional standards and requirements.
- The laws governing pharmacy and drugs.
- Advances in knowledge and technology relating to the use of drugs and public health.

Pharmacists look out for lifestyle problems, disease outbreaks, and other emergency situations in society.[1] Many pharmacists visit institutions such as schools or communities to promote public health. In fact, the current work culture of pharmacists includes some of the following:

- Independent reviews of the treatment.
- Patient advocacy considering best interest of the patient.
- Educate patients about medication.
- Prevent drug-related problems and minimize harm.
- Ensure quality of drugs stored and dispensed and prevent pilferage and drug abuse, etc.

Pharmacists and pharmacies are widely recognized globally through the logo ℞, which is a contraction of an acronym for *recipere* (Latin, "you take") but often written as "℞" in typed text—a command at the beginning of a prescription.[2] An alternative belief is that the ℞ symbol evolved from the Eye of Horus, an ancient Egyptian god supposed to have healing powers.[3] Pharmacies are also recognized through symbols depicting the Bowl of Hygeia (Fig. 2.1), the Greek goddess of hygiene and the daughter of Asclepius (God of Medicine).[4] Other common symbols are the green cross or the mortar and pestle (Fig. 2.2).[5]

Pharmacy is a recognized part of medical practice dating at least as far back as the Sumerian civilization.[6] In ancient times, the physician and pharmacist were the same. The doctor-pharmacist would be needed to possess a thorough

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FIGURE 2.1 Bowl of Hygeia. *From David (developer). https://commons.wikimedia.org/wiki/File:Bowl_Of_Hygieia_by_David.svg.*



FIGURE 2.2 Mortar and pestle with ℞ symbol. *https://commons.wikimedia.org/wiki/File:PharmacistsMortar.svg.*

understanding of the different herbs and plants used as medications. In fact, training in pharmacy was an important component of ancient doctors. In the due course of time, the practice of medicine expanded and became increasingly complicated, necessitating specialization/separation. The first independent pharmacy was opened in Baghdad by the eighth century CE. This separation had formally been made in Europe by 1241 CE to prevent monopolies that could arise from collusion between prescriber and dispenser. Pharmacy being a full-time practice ensures the quality of medicinal plants grown and the drugs dispensed.[7] Compounders (or apothecaries) would prepare solutions, balms, tinctures, etc., extemporaneously (Latin, "at the very time") on prescription from the physician for a particular patient. As the knowledge regarding drugs and drug making became highly specialized and lucrative, more professionals started to enter the pharmacy profession.

In prehistoric times, drugs were mostly plant based. Preserved prescriptions from early civilizations include the following:

- the cuneiform tablets of the Sumerian times (around 2000 to 1500 BCE),
- the Ebers Papyrus (1550 BCE) and Edwin Smith Papyrus (16th century BCE) of the Egyptian civilization,
- the Sushruta Samhita (sixth century BCE) of the Indian Ayurveda, and
- the Shennong Bencao Jing (The Divine Farmer's Herb-Root Classic) compiled during the Han Dynasty China (first century CE).[6]

Most medieval drugs and pharmacy knowledge, however, came from the Greeks and the Romans. One of the pioneer pharmacists was Diocles of Carystus, a Greek physician whose "rhizotomoi" (400 BCE) was an early guide to the uses of medicinal plants. His other works on pharmacotherapeutics were the foundation for Pedanius Dioscorides (first century CE) who went on to write the famous five volume "De Materia Medica" (Concerning Medicinal Substances). It became a basic science for medieval European and Islamic medicine until the 16th century CE.[6]

Perhaps the most important medieval pharmacist was Galen (130−200 CE). Galen's work was instrumental in both pharmacy and medicine through his principles of preparing and compounding drugs by extracting a medicinal agent from plants (roots, rhizomes, leaves, flowers, and bark) and animals by mechanical means with water, alcohol, ether, or fatty

oils. Such medications, called galenicals, include tinctures, infusions, extracts, medicinal waters, alcohols, syrups, soaps, plasters, and liniments. *Because of his contributions to the field, Galen is considered as the father of pharmacy.* These galenicals were the mainstay for nearly 1500 years, and many of his procedures have their counterparts in today's modern compounding laboratories and pharma industries as part of the pharmaceutical formulation.[8] Neogalenicals are the modern counterparts to galenicals, which are highly purified drugs that can be injected as well.[9]

Independent practice of the pharmacy was a revolutionary approach by the Arabs.[10] The Arabs separated the arts of pharmacist (or apothecary) and physician, establishing privately owned drug stores in Baghdad (Baghdad stores) by 750 CE (eighth century CE), which became state regulated by the ninth century CE. During this period, many advances were made in botany and chemistry by pioneers like Muhammad ibn ZakarīyaRāzi (Rhazes) who used chemicals as drugs (865−915) and Abu al-Qasim al-Zahrawi (Abulcasis) who used chemical processes such as sublimation and distillation for the synthesis of complex drugs from simpler chemicals (936−1013).[11] Al-Biruni's (973−1050) Kitab al-Saydalah (The Book of Drugs) and Ibn Sina's Avicenna (The Canon of Medicine) were important writings of this period. These influences reached Europe by the 1200s. In 1240, Frederic II issued a decree to separate the physician's and the apothecary's professions, called the edict of Salerno.[7] The first pharmacy in Europe (still working) was opened in 1241 in Trier, Germany. These pharmacies were called apothecary shops. Fig. 2.3 is published with permission from the New Orleans Pharmacy Museum; it shows the apothecary shop of America's first licensed pharmacist.

The 16th century marked the beginning of the city pharmacopoeias. Although many authors had written manuals/treatises in compounding (formularies), the "Dispensatorium" was the first pharmacopeia published in 1546 by Valerius Cordus under the authority of the Senate of Nuremberg.[12] These pharmacopeias had validity over certain cities only, but they helped delineate the legitimacy for pharmacopeias to be granted by a government or a medical or pharmaceutical society. Ironically, the term pharmacopeia (literally, drug making) was only introduced in 1561 by Anutius Foesius, in his book Pharmacopoeia Medicamentorum Omnium.[13] Other notable city pharmacopeias were the London Pharmacopoeia (first published 1621), the Pharmacopoeia Amstelredamensis (1636), Edinburgh Pharmacopoeia (1899), and the Dublin Pharmacopoeia (1807).

The chemical era started in the 1700s and was the forerunner of modern pharmaceuticals. Swede Carl Wilhelm Scheele (1742−86) made incalculable contributions to kick-start this era by discovering countless organic compounds as well as techniques to isolate organic plant acids. German Friedrich Wilhelm Adam Sertürner (1783−1841) discovered morphine and alkaloids. Pharmaceutical companies such as Merck, Hoffman-La Roche, Burroughs-Wellcome, etc., began as local apothecary shops in the mid-1800s. By the late 1880s, rudimentary methods in organic chemistry had been established by German dye manufacturers, and medicinal chemistry had started.[14] Retail pharmacies started popping up in the United States in 1729. The first attempts at standardizing drugs (or pharmaceuticals) started in 1852 through the foundation of the American Pharmaceutical Association. Instrumental in the association's formation and work was William Procter Jr. (1817−74). For his works, William Procter Jr. is considered as the father of American pharmacy.[15]

The late 1800s gave rise to the biological era and later the antibiotic era.[10] In 1894, Behring and Roux's demonstration of diphtheria antitoxin's effectiveness spurred its production by Parke, Davis & Company (1895) for vaccinating children. For the first time, biological products were used to save thousands of lives. In 1903, Parke-Davis received US Biological License No. 1. In 1955, poliomyelitis vaccine was produced. Another important event was Alexander Fleming's discovery of penicillin. This era also gave birth to modern pharmacology through the invention of the technique of drug discovery: screening for chemicals causing particular functional activity followed by receptor identification and biochemical understanding of involved protein and finally using medicinal chemistry to optimize a good drug.



FIGURE 2.3 Apothecary shop. *Published with permission from New Orleans Pharmacy Museum.*

Similar principles were used to optimize biological drugs. Biological drugs can help with a lot of different conditions in the body, such as monoclonal antibodies to fight cancer or filgrastim to boost white blood cells for a patient undergoing chemotherapy. The biosimilar is a generic version of the innovator biological drug.

Modern national pharmacopeias came out in the 1800s. By the late 1800s, it was realized that national agencies need to regulate pharmacopeias within the country. The British Pharmacopoeia[16] was the first national pharmacopeia. Although not the most popular at the onset, it paved the way for quality control in drug manufacturing. Similarly, the United States Pharmacopoeia[17] was another important pharmacopeia, which has become accepted worldwide. The International Pharmacopoeia is published by the World Health Organization (WHO) for adoption by the Member States and to help achieve a potentially global uniformity of quality specifications for selected pharmaceutical products, excipients, and dosage forms.[18]

The late 1950s observed a split of the pharmacy profession into the industrial pharmacy (pharmaceutical industry) and clinical pharmacy.[19] By the 1950s, as a result of advancements described above, the industrial pharmacy had replaced compounding pharmacy through standardized manufacturing of drugs (and other pharmaceuticals). Large-scale manufacturing of medicinal products in the pharmaceutical industry and the introduction of prescription-only legal status for most therapeutic agents limited the role of pharmacists to compounding. Pharmacies became retail outlets of pharmaceutical industries. This shift urged pharmacists to focus more on patient (clinical) services.

Clinical pharmacy, specifically hospital pharmacy, arose as a response to the rise of industrial pharmacy.[19] Until the 1960s, pharmacists were mostly product experts who compounded drugs. By the mid-1960s pharmacists had evolved toward a more patient care-oriented practice and developed the concept of providing clinically relevant expertise, especially in hospitals. It marked the beginning of a period of rapid transition that was characterized by closer interaction with physicians and other healthcare professionals.

The Ninth-Floor Pharmacy project delineated clinical pharmacy practice in the modern age.[20] With the rise in hospitals and improved needs for safe drug use, pharmacists began facing many difficulties managing drug distribution and dosage in the hospital pharmacies. The Ninth-Floor Pharmacy project, headed by William Smith at the University of California, San Francisco (UCSF) School of Pharmacy, was started at UCSF Medical Center on September 7, 1966. The project was to develop a hospital floor-based hospital pharmacy service that provided optimized drug use in hospitalized patients. In essence, the project created the modern clinical pharmacist by outlining their roles in patient care (provide and disseminate drug information, train pharmacy students, design and conduct studies in cooperation with clinicians within the framework of a team approach to patient care). The project also opened the way for drug information service and drug and therapeutics committees in hospitals.

The increased role of clinical pharmacists led to the pharmaceutical care era.[21] Hospital pharmacies eventually led to the birth of various forms of clinical pharmacy such as ambulatory care pharmacy, oncology pharmacy, pediatric pharmacy, pharmacotherapy, etc. Other pharmacies continued in operation as community pharmacy, veterinary pharmacy, mail-order pharmacy, or internet pharmacy. Developments in ambulatory care opened scopes for community pharmacists to provide pharmaceutical care services that used to be provided by clinical pharmacists. In 1990, Hepler and Strand and the WHO introduced the concept of pharmaceutical care as it was increasingly realized and established that pharmacists have an important role in patient care. Furthermore, pharmacists individually and as a profession have important roles to play in positively influencing drug policy, drug use, and outcomes, as well as other aspects of healthcare. In many instances, this will be through collaboration with other health professionals at a community level. Notably, pharmaceutical care has evolved in the United States as medication therapy management,[22] which is medical care provided by pharmacists aimed primarily at optimizing drug therapy and improving therapeutic outcomes for patients. An important development arising from pharmaceutical care is that of good pharmacy practice (GPP) to provide optimal evidence-based care. Another field that has emerged is pharmacovigilance. Pharmacovigilance constitutes safety monitoring of drugs (such as early detection of new adverse drug reactions and other drug-related problems) and is rapidly becoming an integral part of pharmaceutical care. Education and training of healthcare professionals on drug safety and effectiveness improved therapeutic outcomes. All these developments and activities have become more important as consumer-driven, not consumer-oriented practice; pharmaceutical care has emerged with consumers becoming more active, independent, informed, critical, and reflective of care.

Advancement in life sciences and pharmaceutical sciences has pushed pharmaceuticals and pharmacy practice to the era of genomics and personalized medicine/personalized pharmacy in the last three decades.[23] The omics revolution, especially genomics, has opened new horizons in healthcare. With genomics, the parts of the genome and the proteins responsible for diseases can be determined. It opens up the possibility of creating drugs that affect the genome and proteins, and more so in a tailored or personalized manner for each patient. Some drugs are indicated after quick genetic tests in the patients. Many pharmacists practice personalized pharmacy, especially in oncology. Although incredibly powerful, genomic drugs raise

concerns of ethics and privacy. The integration of nanotechnology, biotechnology, and genomics in pharmaceuticals has further created new opportunities, exciting potential, and newer challenges to the pharmacy profession.

Many changes in pharmacy practice due to innovations in other fields have also occurred in the last few decades. Perhaps the most noticeable change is due to IT and related technologies. A case in point is the rise of internet pharmacies where one can purchase many drugs without prescription in most cases. Other changes are in the supply chain, with the use of pharmacy robots, smart packaging, and a "smart pill." Smart packaging is a blister pack containing a microchip that can monitor when doses are popped out of the package, with data transmitted to a mobile phone or tablet app. Because smart packaging cannot tell if the dose was taken, a smart pill is developed that has sensors that can provide information on dose, heart rate, and other variables once ingested by the patient. *A curious but not surprising case is the return of extemporaneous prescription compounding* for special cases such as dental care or customizing formulations or preparing drugs in new forms (e.g., a medicated lollipop containing the drug for patients who have difficulty in swallowing). Electronic prescribing (EP) systems automate prescribing, supply, and administration of drugs in hospitals, where they have been shown to reduce medication errors and have a high impact on patient safety. Barcode identification of drugs has been used with EP systems and has been shown to reduce medication administration errors, as well as improve the completeness of the medication history. In the United Kingdom, the Falsified Medicines Directive (2017) calls for unique identification of drugs at the point of dispensing to combat counterfeiting.[24] Telecare[25] and Remote Patient Management[26] are other technology applications in the pharmacy that use digital communications technology (audio and visual) to provide healthcare consultations and services and monitoring of patients remotely.

OBJECTIVE 2.2. OVERVIEW ERAS OF PHARMACEUTICAL INDUSTRIES AND THEIR JOBS FOR PHARMACY GRADUATES

The pharmaceutical industry had its origin in pharmacies and chemical companies.[14] The modern pharmaceutical industry traces its origin to

- apothecaries (Germany: Merck, Schering; Switzerland: Hoffmann-La Roche; United Kingdom: Burroughs-Wellcome; France: Etienne Poulenc; United States: Abbot, Smith Kline, Parke-Davis, Eli Lilly Squibb, Upjohn) that moved into wholesale production of drugs such as morphine, quinine, and strychnine in the middle of the 19th century, and
- dye and chemical companies (Agfa, Bayer, and Hoechst in Germany; Ciba, Geigy, and Sandoz in Switzerland; Imperial Chemical Industries in England; and Pfizer in the United States) that established research labs and discovered medical applications for their products starting in the 1880s.

A merger of these two types of firms took place with the emergence of pharmaceutical chemistry and pharmacology at the end of the 19th century. The remarkable thing was that pharmaceutical firms, first in Germany in the 1880s and later in the United States and England, established cooperative relationships with academic labs into a massive industrial research program that continues to the present.

Chemical era. By the early 19th century, chemists were able to extract and concentrate traditional plant-based remedies, giving rise to treatments such as morphine and quinine. By the early 20th century, similar methods applied to animal-based remedies resulted in the isolation of epinephrine, which was the first hormone to be used as a drug. German dye manufacturers, meanwhile, had already developed synthetic organic chemistry as an industrial discipline, especially in deriving dyes from coal tar. Chemists were soon able to modify raw dyes and their by-products to make them more effective as drugs, effectively beginning pharmaceutical and medicinal chemistry. One of the first chemically synthesized drugs, aspirin (1897) by the German pharmaceutical company Bayer, continues to be used and researched today.

Biological era and the beginning of regulation and mass production. The end of the 19th century also witnessed the development of several important vaccines, including those for tetanus and diphtheria. Similarly, the antisyphilitic organoarsenic compound salvarsan was developed as one of the first antimicrobial chemotherapeutic agents. In the early 20th century, the progressive role of chemical and life sciences began appearing. World War I blockades forced US chemists to replicate German processes for producing drugs such as aspirin, salvarsan, and veronal. As a result, the focus shifted to large-scale synthesis and newer drug discovery, including biologicals (or biologics) and small molecules. At the same time, regulations and laws began appearing to regulate pharmaceuticals. Regulations against secret recipes and fallacious claims were institutionalized. Professional bodies of physicians and pharmacists and national formularies set manufacturing standards. The development of diphtheria antitoxin in the 1890s and associated issues of inactive antitoxin forced the 1902 Biologics Control Act. The 1906 Food and Drugs Act in the United States and similar laws in several other countries prohibited adulteration. Manufacturers have to reveal ingredients on product labels.

After the expiry of the patent, generic versions of the drugs could be produced and marketed by any pharmaceutical companies. Patent expiry follows production by different manufacturers and improved access to drugs as the generic versions.

The biologics market is growing rapidly. Several biologics have sales of more than $1 billion annually. For example, in 2011, global sales were $7.19 billion for Remicade (infliximab) and $5.98 billion for Avastin (bevacizumab). In addition, many of these drugs when out of patent *biosimilars* (generic version of biologics) enter the market, e.g., Herceptin (trastuzumab), Humalog (insulin lispro), Rituxan (rituximab [MabThera in Australia]), Remicade, and Aranesp (darbepoetin alfa).[27,28] Furthermore, the opportunity exists, but unlike the generic market where market entry is fairly simple and well established, there are substantial barriers to entry into the biosimilar market.

Genomic era. Following an explosive growth, the pharmaceutical industry, which is always innovating, began looking at newer ways of therapy. With a better understanding of the disease and a more patient-centered approach, newer drugs, diagnostics, and lines of research emerged. The understanding of cellular processes, including the interaction of genes, proteins, and cellular machinery has given rise to targeted therapy and genomic drugs. An example is Benlysta (belimumab), which is a result of research through genomics and bioinformatics, and is the first new therapy for systemic lupus erythematosus in 50 years. Similarly, with the understanding that multiple cellular targets may be responsible for various forms of breast cancer, different targeted therapy is given that has transformed prognosis in breast cancer. Epigenetics has helped us target expression or silencing of select non-DNA genetic elements (such as siRNA, proteins) to change disease manifestation. Stem cell therapies, synthetic biology, and tissue engineering are providing newer approaches to treating diseases. Concepts like network medicine are improving understanding of the molecular relationships between apparently distinct manifestations of a disease. The pharmaceutical industry is undergoing an exciting revolution in the field of genomics.

Comparison of Financial Volume of Pharmaceutical Industries With Other Industries

The worldwide market for pharmaceuticals was projected to grow from around $1 trillion in 2015 to $1.3 trillion by 2020, representing an annual growth rate of 4.9%.[29] Several global demographic and economic trends are driving pharmaceutical consumption, including a rapidly aging world population and an associated rise in chronic diseases, increased urbanization and higher disposable incomes, greater government expenditure on healthcare, and growing demand for more effective treatments. The United States is a large pharmaceutical's market with a huge market size.[29] Developing country markets are relatively more heterogeneous with groups of high purchase power that spend more on innovation and groups of low purchase power that spend more on generics. Top markets for pharmaceutical products continue to be developed countries in North America, Western Europe, and East Asia with high per capita spending on healthcare, growing elderly populations, and advanced regulatory systems. Though ranked lower, there are growing opportunities in developing countries such as China and India as incomes and healthcare spending increase.

The pharmaceutical industry is known for its competitiveness, large employment, directly and indirectly, innovation as one of the most R&D-intensive industries (15%−20% of revenues to R&D activities in the United States), and fast-growing segment of biologics and generics.[29]

Global pharmaceutical, biotech, and medtech markets have shown a rising trend.[30] Although the pharmaceutical market is large, it is still a fraction of the chemical industry.[31]

Career Opportunities for Pharmacy Graduates in Pharmaceutical Industries

Pharmacy graduates have many career options in pharmaceutical industries. Traditionally, pharmacists have used their clinical knowledge in a variety of practice settings, including community pharmacies and hospitals. However, there are also many significant, alternative career opportunities within the pharmaceutical and biopharmaceutical industries. With advances in medical technology, the pharmaceutical and biopharmaceutical industries are constantly expanding efforts to discover, develop, and market new drugs, thereby creating more employment opportunities for pharmacists. The pharmacy professionals working on drug innovation and research are called pharmaceutical scientists. Some of the job areas in pharmaceutical industries include the following:

- clinical trials,
- pharmacovigilance,
- pharmaceutical formulation development,
- manufacture and supply,

- quality assurance,
- regulatory affairs,
- medical affairs,
- pharmacokinetics and drug metabolism,
- health economics and outcomes research,
- project management,
- commercial business development, and
- marketing and sales.

OBJECTIVE 2.3. DISCUSS POLICIES AND LAWS GOVERNING THE PHARMACY PROFESSION AND PUBLIC HEALTH

World Health Organization

International health policies often have influence to standardize and regulate the pharmacy profession. In this regard, policies set by the WHO have been significant in shaping pharmacy and other health professions globally. One of the most important international regulations set by the WHO has been the *right to health*, which was evident since the proposal of the WHO Constitution 1946 and the Universal Declaration of Human Rights 1948. The binding International Covenant on Economic, Social, and Cultural Rights of 1966 detailed realization of the right to health through bettering access to health facilities, goods, and services. The WHO introduced the concept of *Essential Medicines and National Drug Policy* in 1977 after the World Health Assembly 1975 in an attempt to make good quality modern drugs available and affordable.[32] National drug policies for ensuring essential drugs in member countries were considered to improve pharmaceutical care. Furthermore, the WHO in 1978, through the Declaration of Alma Ata, identified *provision of essential drugs* as one of the eight elements of primary healthcare. Declaration of Alma Ata by the WHO is one of the first of its kind to emphasize primary care.[32] Together, provision of essential drugs and national drug policies and various other WHO policies related to these have improved access to drugs and care in various WHO member countries. Going a step further, in 2000, WHO released the *WHO Model Formulary* to promote safe and cost-effective use of drugs and *WHO Model Quality Assurance (QA) System* for prequalification of vendors especially for UN agencies. The model formulary and model QA system provided important guidance to member countries developing their own formularies and QA systems. In 2000, the authoritative General Comment 14 (2000) also applied the principles of accessibility, availability, appropriateness, and assured quality to goods and services, which include essential drugs, to the right to health.[33] Through drug and therapeutic guidelines in association with the International Network for Rational Use of Drugs, rational prescribing guidelines have been released by the WHO.[34]

The International Pharmaceutical Federation (FIP) and WHO together developed GPP guidelines, first released in 1996, which ensure quality in pharmaceutical care.[35] In 1999, WHO released the *Counterfeit Drug Policy* guidelines to help member countries combat counterfeit drugs.[36] In the same year, the International Conference on Harmonisation (ICH) and WHO also released *Good Manufacturing Practice* (GMP): a system for ensuring that products are consistently produced and controlled according to quality standards minimizing the risks involved in any pharmaceutical production that cannot be eliminated through just testing the final product but by testing each step in the manufacturing process. Many countries have formulated their own requirements for GMP based on WHO GMP and others have harmonized their requirements through the Pharmaceutical Inspection Convention.[37,38]

WHO also improved health security through the International Health Regulations (IHR) 2005, a binding document between 196 countries including all WHO Member States.[39] IHR has been used as a guiding principle for many countries[40] and for coalitions like Global Health Securities Agenda for implementing health security, including pharmaceutical availability and stocking, for health emergencies and outbreaks.[41]

Apart from this, WHO has played a crucial role in drug utilization statistics with the aim of improving drug use. For this, WHO has defined the Anatomical Therapeutic Chemical (ATC[42]) system under which defined daily dose (DDD[43]), an assumed average maintenance dose per day for a drug used for its main indication in adults, is assigned. DDDs are assigned for drugs that are approved in at least one country. DDDs are not assigned for radiopharmaceuticals, appetite stimulants, anesthetics, blood substitutes, parenteral organic nitrates, many topical products, vaccines, antineoplastic agents, etc., due to high dose variation.[44] In the ATC classification system, the active substances are divided into different groups according to the organ or system on which they act and their therapeutic, pharmacological, and chemical properties. Like the International Classification of Diseases (ICD) that assign a unique codes for diseases, the ATC codes are unique to drugs based on anatomical therapeutic and chemical classification. It is possible for a drug to have more than one ATC

code if it has more than one approved therapeutic indication. The ATC/DDD system is not exactly a therapeutic classification system and is for statistical studies regarding drug safety.[45]

WHO has pushed for standardization in the naming of drugs through the use of International Nonproprietary Names[46,47] (INN or generic names) globally over brand names since 1950. INN program gives a unique name to pharmaceutical substances or active pharmaceutical ingredients, which is globally recognized and is public property. INN is meant for pharmacopoeias, labeling, product information, etc. There were around 7000 names on the INN list in 2017, growing by some 120−150 names every year. INN has made naming of NDEs (new drug entities) that have complex IUPAC chemical names easier as the INN nomenclature follows a system for naming based on the IUPAC name and chemical structure, Chemicals Abstract Service registry number, as well as checking with previously existing INNs. INNs are published in the *WHO Drug Information*.

United States Drug Legislations

The US federal legislation—especially the FDC Act 1938, CSA 1970, PPMA 1970, PDMA 1970, OBRA-90, and HIPAA—has shaped pharmacy practice in the United States and globally.[48] Most of the federal legislation has been initiated in response to issues and concerns at certain points in time to protect the health, safety, and welfare of the patient from the potential risks of drug use or misuse. Most legislation has also imposed new requirements in such areas as record-keeping, counseling, and packaging of pharmaceuticals.

The Food, Drug, and Cosmetic (FDC) Act of 1938 provides the basis of quality control of drugs and penalties for spurious drugs. FDC Act of 1938 served as a replacement for the Food and Drug Act of 1906, which did not prohibit false therapeutic claims and, in some cases, even protected those claims. The 1906 law did not enforce labeling list ingredients, directions for use, or provide warnings. In 1937, more than 100 people, many of whom were children, died after a sulfa drug was administered that did not have proper labeling. As a result, FDC 1938 was passed, which brought cosmetics and medical devices under control and required proper labeling. It also mandated premarket approval of all new drugs, such that a manufacturer would have to prove that a drug was safe before it could be sold. FDC Act ensures the quality and safety of pharmaceuticals and regulates dispensing by the Durham−Humphrey Amendment to it (1951).[7] The FDC Act penalizes violations regarding drug products,[8] especially counterfeit or adulterated or misbranded drugs: the holding, dispensing, and sale of which are in violation of the Act.

The Comprehensive Drug Abuse Prevention and Control Act of 1970 or Controlled Substance Act (CSA) provides the basis against abuse of drugs. The CSA exerts more of a direct impact, as the law requires registration, specific record keeping, and rules regarding the dispensing of controlled substances. Penalties for the violation of the CSA are significantly more severe than a violation of the FDC Act.

The Poison Prevention Packaging Act (PPPA) of 1970 ensures safe packaging. The enactment of PPPA in 1970 reduced the oral prescription drug-related death rate of 1.4 deaths per million children under 5 years of age due to child-resistant packing, a reduction of up to 45%. Failure to comply with packaging requirements or any of the applicable regulations is considered a misbranding violation under the FDC Act.

The Prescription Drug Marketing Act of 1987 (PDMA), incorporated into the FDC Act, addresses the diversion of drugs into a secondary gray market commonly happening through drug-marketing practices. These marketing practices—including the distribution of free samples and the sale of deeply discounted drugs to hospitals and healthcare entities—have helped fuel a multimillion-dollar drug diversion market that provides a portal through which mislabeled, subpotent, adulterated, expired, and counterfeit drugs can enter the nation's drug distribution system. PDMA prohibits the act or offer of knowingly selling, purchasing, or trading a prescription drug sample. Another important portion of this extensive law that affects pharmacists prohibits the resale of any prescription drug that was previously purchased by a hospital or other "healthcare entity."

The Omnibus Budget Reconciliation Act of 1990 (OBRA-90) places expectations on the pharmacist in how to interact with the patient. Although the primary goal of OBRA-90 was to save the federal government money by improving therapeutic outcomes, the method to achieve these savings was implemented by imposing on the pharmacist counseling obligations, prospective drug utilization review (ProDUR) requirements, and record-keeping mandates. As part of the ProDUR, the following are areas for drug therapy problems that the pharmacist must screen for therapeutic duplication, drug−disease contraindications, drug−drug interactions, incorrect drug dosage, incorrect duration of treatment, drug−allergy interactions, and clinical abuse/misuse of medication. OBRA-90 overall resulted in standardized pharmaceutical care to all patients, although it was initially meant for Medicaid (one of the US health insurance program) patients.

The Health Insurance Portability and Accountability Act of 1996 (HIPAA) has been safeguarding the privacy of protected health information of patients since 2003. Pharmacies that maintain patient information in electronic format or

conduct financial and administrative transactions electronically, such as billing and fund transfers, must comply with HIPAA. Although HIPAA places stringent requirements on pharmacies to adopt policies and procedures relating to the protection of patient protected health information (PHI), the law also gives important rights to patients. These rights include the right to access their information, the right to seek details of the disclosure of information, and the right to view the pharmacy's policies and procedures regarding confidential information. Typically, HIPAA restricts transmission and disclosure of PHI and strict adherence of the pharmacists/pharmacies to HIPAA. The pharmacy may disclose PHI to "business associates," which are billing services, claims processing, utilization review, or data analysis through stringent contractual agreements. HIPAA has security standards designed to protect the confidentiality of PHI that is threatened by the possibility of unauthorized access and an interception during electronic transmission.

HIPAA, through its different rules, indirectly forces healthcare organizations to comply with legislation. It is because HIPAA essentially enforces that all healthcare entities and organizations that use, store, maintain, or transmit patient health information ensure privacy, reduce fraudulent activity, and improve data systems. Accordingly, data regarding the use of drugs and health products are monitored by agencies to ensure use of drugs and health products that conform to FDC, CSA, PPMA, PDMA, and OBRA-90. By preventing security risks that could result in major compliance costs, HIPAA makes organizations focus on quality in drugs and health products as noncompliance results in massive fines.[49] Many countries have developed their policies and have taken inputs from the WHO and US policies and regulations (Table 2.1).

TABLE 2.1 An Overview of the Different Laws Governing Pharmacy Practice in Select Countries: United States,[48] United Kingdom,[50] Ireland,[51] India, and Nepal[52]

United States	United Kingdom	India
Pure Food and Drug Act of 1906 Harrison Narcotics Tax Act of 1914 Food, Drug, and Cosmetic Act of 1938 Comprehensive Drug Abuse Prevention and Control Act of 1970 (with updates) Occupational Safety and Health Act of 1970 Poison Prevention Packaging Act of 1970 Medical Device Amendments of 1976 Federal Anti-Tampering Act of 1983 Orphan Drug Act of 1983 Drug Price Competition and Patent-Term Restoration Act of 1984 Prescription Drug Marketing Act of 1987 Omnibus Budget Reconciliation Act of 1987 Anabolic Steroids Control Act of 1990 Omnibus Budget Reconciliation Act of 1990 Dietary Supplement Health and Education Act of 1994 Uruguay Round Agreements Act of 1994 Health Insurance Portability and Accountability Act of 1996 Food and Drug Administration Modernization Act of 1997 2002 Best Pharmaceuticals for Children Act Medicare Modernization Act of 2003 Combat Methamphetamine Epidemic Act of 2005 Medicaid Tamper-Resistant Prescription Law of 2007 Food and Drug Administration Amendments Act of 2007 Elements of risk evaluation and mitigation strategies Restricted drug programs Thalidomide Clozapine Buprenorphine Isotretinoin Patient Protection and Affordable Care Act of 2010 Durham–Humphrey Amendment of 1951 Kefauver–Harris Amendment of 1962	Pharmacy Order 2010 (Statutory Instrument 2010 No. 231) and the various sets of statutory rules made under the Order Medicines Act 1968 Poisons Act 1972	Drug and Cosmetic Act,1940 and Rules, 1945 The Narcotics Drug and Psychotropic Substances Act and Rules, 1985 Drug Price Control Order, 1995 Consumer Protection Act, 1986 Infant Milk Substitutes, Feeding Bottles, and Infant Food Acts, 1992 Drug and Magic Remedies Act and Rules, 1954 Prevention of Food and Adulteration Act, 1954 The Pharmacy Act, 1948 The Drugs and Magic Remedies (Objectionable Advertisement) Act, 1954
	Ireland	**Nepal**
	Pharmacy Act 2007 European Communities Regulations 2008 European Union Regulations 2017 Regulation of Retail Pharmacy Businesses Regulations 2008 Health (Pricing and Supply of Medical Goods) Act 2013 Health Act 2007 (Care and Welfare of Residents in Designated Centres for Older People) Regulations 2013 Irish Medicines Board Act 1995 Medicinal Products Regulations 2007 Misuse of Drugs Regulations 1988	Drug Act, 1978 Nepal Pharmacy Council Act, 2000 Narcotic Drugs (Control) Act, 1976 Code for Manufacture of Drugs, 1986 Code for Sale and Distribution of Drugs Code for Advertisement of Drugs National Drug Control Policy, 1995 Hospital Pharmacy Service Act

International Pharmaceutical Federation or Fédération Internationale Pharmaceutique

FIP is the largest international pharmaceutical society that regularly contributes to the advancement of the pharmacy profession. Founded in 1912, FIP is a nongovernmental organization with its head office in the Netherlands. FIP coordinates observance of World Pharmacists Day on September 25th every year.[53] In 1912 the FIP was founded on September 25th.

It is important that modern pharmacists understand their increasingly complex role in light of complicated patient care guidelines, public health guidelines, a multitude of pharmaceuticals, and legal requirements. To aid in this process, the GPP guidelines and code of ethics (CoE) for pharmacists serve as a suitable starting point to provide professional pharmacy care.

Good Pharmacy Practice

Pharmacists are charged by their national or other appropriate (e.g., state or provincial) authorities with the management of the distribution of drugs to consumers and to engage in appropriate efforts to assure their safe and efficacious use. Not only that, pharmacists are increasingly accepting greater responsibility for the outcomes of drug use and are evolving their practices to provide patients with enhanced drug-use services. Increasingly complex and diverse nature of pharmacists' roles in the healthcare system demands that pharmacists follow GPP guidelines.[35] According to FIP/WHO, GPP is "the practice of pharmacy that responds to the needs of the people who use the pharmacists' services to provide optimal, evidence-based care. It is essential that there be an established national framework of quality standards and guidelines."

- A pharmacist's first concern in all settings is the welfare of patients;[54]
- The core of pharmacy activity is to help patients make the best use of drugs. Fundamental functions include the supply of medication and other healthcare products of assuring quality, the provision of appropriate information and advice to the patient, administration of medication, when required, and the monitoring of the effects of medication use;
- An integral part of the pharmacist's contribution is the promotion of rational and economic prescribing, as well as dispensing;
- The objective of each element of pharmacy service is relevant to the patient, is defined, and is effectively communicated to all those involved. Multidisciplinary collaboration among healthcare professionals is the key factor for successfully improving patient safety.

Necessary conditions to meet the GPP guidelines.[54] The following conditions are deemed necessary:

- The well-being of patients should be the main philosophy underlying practice, even though it is accepted that ethical and economic factors are also important.
- Pharmacists should have input into decisions about the use of drugs. A system should exist that enables pharmacists to report and to obtain feedback about adverse events, medication errors, misuse or drug abuse, defects in product quality, or detection of counterfeit products. This reporting may include information about drug use supplied by patients or health professionals.
- The relationship with other health professionals, particularly physicians, should be established as a therapeutic collaborative partnership that involves mutual trust and confidence in all matters relating to pharmacotherapy.
- The relationship between pharmacists should be one of the colleagues seeking to improve pharmacy service, rather than acting as competitors.
- In reality, organizations, group practices, and pharmacy managers should accept a share of responsibility for the definition, evaluation, and improvement of quality.
- The pharmacist should be aware of essential medical and pharmaceutical information (i.e., diagnosis, laboratory test results, and medical history) about each patient. Obtaining such information is made easier if the patient chooses to use only one pharmacy or if the patient's medication profile is available.
- The pharmacist needs evidence-based, unbiased, comprehensive, objective, and current information about therapeutics, drugs, and other healthcare products in use, including potential environmental hazards caused by disposal of drugs' waste.
- Pharmacists in each practice setting should accept personal responsibility for maintaining and assessing their competence throughout their professional working lives. Although self-monitoring is important, an element of assessment and monitoring by the national pharmacy professional organizations would also be relevant in ensuring that pharmacists maintain standards and comply with requirements for continuous professional development.
- Educational programs for entry into the profession should appropriately address both current and foreseeable changes in pharmacy practice.

- National standards of GPP should be specified and should be adhered to by practitioners.

Frameworks needed to implement GPP.[54] For GPP implementation, three frameworks are needed at the national or appropriate (e.g., state or provincial) level:

- A legal framework that ensures the integrity of supply chain and quality of drugs; defines who can practice pharmacy; and what the scope of pharmacy practice is.
- A workforce framework that defines personnel needed and ensures the competence of such staff through continuing professional development programs.
- An economic framework that provides sufficient resources and incentives to ensure the activities undertaken in GPP.

Code of Ethics in Pharmacy Practice

Laws and ethics of healthcare overlap considerably because both share the concern that the conduct of healthcare professionals should reflect respect for the well-being, dignity, and self-determination of the public. At the same time, there are situations in which the two domains of law and ethics may remain distinct and a CoE, while preparing with awareness of the law, is addressed to ethical obligations. Laws shall have their loopholes, and it cannot ensure fair practices without a CoE.

Although the specifics of CoE vary slightly country by country (influenced by cultures), most of them define and seek to clarify the obligations of the pharmacists to use their knowledge and skills for the benefit of others, to be fair and just in their service to the public, to minimize harm, and to respect patient autonomy. The CoE educates pharmacists about their ethical duties and obligations and serves as a tool for guiding pharmacists in relationships with patients, health professionals, and society, as well as for continued competence (via continuing education), self-evaluation, and peer review. Perhaps the best international CoE is that of points from FIP CoE (Table 2.2).[55]

Of course, the FIP CoE is more a guideline than a statute. Different countries and associations adopt their own CoEs.

International Patient Safety Goals

Joint Commission International accredited healthcare institutions insist on International Patient Safety Goals to improve patient safety.[65] The six goals are:

Goal 1: Identify patients correctly
Goal 2: Improve effective communication

TABLE 2.2 International Pharmaceutical Federation (FIP) Code of Ethics (2014)

FIP Code of Ethics for Pharmacists (2014)
• to act with honesty and integrity in their relationships with consumers, patients and carers, and other health professionals, including pharmacy practice colleagues, and not engage in any behavior or activity likely to bring the profession into disrepute or to undermine public confidence in the profession;
• to ensure that their priorities are the safety, well-being and the best interests of those to whom they provide professional services and that they act at all times as autonomous health professionals, recognizing the challenges posed by dividing loyalties and the potential in many settings for conflicts of interest that need careful management;
• to always act professionally, in accordance with scientific principles and professional standards, including those developed by the international pharmaceutical federation;
• to cooperate and collaborate with colleagues, other health professionals, consumers, patients, carers and other actors in the healthcare delivery system to ensure that the best possible quality of healthcare is provided both to individuals and the community at large, while always considering the limitations of available resources and the principles of equity and justice;
• to respect and protect the confidentiality of patient information acquired or accessed in the course of providing professional services and to ensure that such information is only disclosed with the informed consent of that individual or as allowed by applicable legislation and regulation;
• to respect patients' rights and recognize and respect the cultural differences, beliefs, and values of patients, carers and other healthcare professionals particularly in the event of a conflict with their own moral or religious beliefs;
• to ensure continuity of care for the patient in the event of a conflict with their own moral or religious beliefs, based on respect for patient autonomy;
• to comply with legislation and accepted codes and standards of practice in the provision of all professional services and pharmaceutical products and to ensure the integrity of the supply chain for drugs; and
• to ensure that they maintain competence through continuing professional development.

Goal 3: Improve the safety of high-alert medications
Goal 4: Ensure safe surgery
Goal 5: Reduce the risk of healthcare-associated infections
Goal 6: Reduce the risk of patient harm resulting from falls.

These patient safety goals are relevant to pharmacists in at least the following ways. Goal 1: Identifying patients correctly is essential to provide the right drug to the right patient. Goal 2: Improved effective communication focuses on patient counseling and medication reconciliation and to improve medication adherence. Goal 3: Improve safety of high-alert medication and consider other approaches to prevent adverse medication outcomes, especially of certain medications with expected immediate life-threatening events, e.g., adrenaline, insulin, etc. Goal 4: Ensure safe surgery concerns with proper use of anesthetics. Goal 5: Reduce the risk of healthcare-associated infections related to antimicrobial stewardship. Goal 6: Reduce the risk of patient harm resulting from falls as many drugs as associated with drowsiness.

OBJECTIVE 2.4. DETAIL AN OVERVIEW OF PUBLIC HEALTH AND DISEASE PREVENTION

Public health is the science and systems designed to protect and improve health at the community and population level. It includes efforts to improve health by preventing disease and promoting healthy behaviors.[56] These efforts include initiatives to educate the public about healthier choices, facilitating healthier lifestyles through community support, preventing outbreaks and the spread of infectious diseases, and ensuring safe food and water in communities. Public health includes the practices, procedures, institutions, and disciplines required to improve population health. Although the pharmacy practice needs to consider the impact it has on public health constantly, it must be increasingly recognized that public health activities, locally or globally, influence the decisions, roles, and scope of pharmacists. Therefore, pharmacists need to understand the public health picture and players to fulfill their increasingly complex role in maintaining individual and public health. Particular focus must be placed on the organizations involved in international public health as these organizations have affected local and national level public health initiatives and activities.

Organizations Involved in International Public Health

Apart from WHO, a large number of organizations of various sizes are involved in international public health. These are involved in either providing long-term healthcare or providing special aid to victims of war, famine, and natural disasters. Many have overlapping and crosscutting programs.

WHO has been the forefront organization in international public health since 1947.[57,58] WHO is an intergovernmental agency related to the United Nations, and its principal goal is "the attainment by all peoples of the highest possible level of health." WHO directs and coordinates international health activities by supplying technical assistance to member countries. It develops norms and standards, disseminates health information, promotes research, provides training in international health, collects and analyzes epidemiologic data, and develops systems for monitoring and evaluating health programs. WHO's scope of work includes health systems; noncommunicable diseases; communicable diseases; promoting health through life course; preparedness, surveillance, and response during emergencies; and corporate services to provide the enabling functions, tools, and resources for conducting its programs. An overview of the public health activities done by WHO is given in Table 2.3.

Organizations Providing Long-Term Healthcare.[60,61] The *World Bank* is another major intergovernmental agency related to the UN heavily involved in international health. The World Bank loans money to developing countries on advantageous terms not available in commercial markets, especially in health, nutrition, and education.

UN agencies such as UNICEF, UNFPA, and UNDP also provide public health aid. UNICEF focuses on children's healthcare (mostly in cooperation with WHO), especially that of the world's most vulnerable children younger than 5 years of age. UNICEF also provides aid on water supply and sanitation, child nutrition, and emergency relief. UNFPA spends on family planning programs, especially in high population growth and high poverty rate countries. UNDP spends on health, education, and employment. Its focus is on AIDS, maternal and child nutrition, and excessive maternal mortality. UNAIDS works for HIV/AIDS-related programs. UNITAID works on pharmaceutical supply chain programs in tuberculosis (TB), malaria, hepatitis C, etc. The Global Fund for AIDS, Malaria, and Tuberculosis works across crosscutting programs of AIDS, malaria, and TB in developing countries.

TABLE 2.3 WHO and Global Health: Historic Landmarks

1945	1946	1947	1948	1950
Diplomats decide on the need for a global organization overseeing global health and plan for the creation of WHO in San Francisco	WHO's constitution is drafted and then approved at the International Health Conference in New York City	WHO establishes the first ever global disease-tracking service, with information transmitted via telex	WHO's Constitution ratified on April 7—"World Health Day." Next two decades are focused on mass campaigns for communicable diseases[a]	WHO begins advising countries on the responsible use of antibiotics
1952–57	1963	1969	1972	1974
Mass global vaccination campaigns facilitation for polio eradication	Measles vaccination begins followed by mumps and rubella during the next 6 years	1st International Health Regulations are established	HRP[b] program is to research sexual and reproductive health and rights	WHO founds the Expanded Program on Immunization
1975	1977	1978	1987	1988
WHO founds TDR[c] program to combat diseases of poverty	The first essential drugs list is published	Alma Ata convention[d] sets the aspirational goal "Health for All" emphasizing Universal Health Coverage (UHC)	The first antiretroviral medication is licensed, prompting a shift in WHO's priorities to HIV/AIDS treatment	The Global Polio Eradication Initiative is established
1995	1999	2000	2001	2003
The DOTS strategy for reducing the toll of tuberculosis (TB) is launched	The Global Alliance for Vaccines and Immunization is formed	MDGs[e] are formed with specific goals for health. The WHO Global Outbreak Alert and Response Network is established	The Global Fund to fight AIDS, Tuberculosis, and Malaria is created	WHO Framework Convention on Tobacco Control is ratified. WHO launches the "3 by 5" initiative[f]
2004	2005	2006	2008	2009
Indian Ocean tsunami disaster: the Strategic Health Operations Centre is used for the first time to assist with global emergency coordination	The International Health Regulations are revised	The number of children who die before their fifth birthday declines below 10 million for the first time in recent history	The World Health Statistics report notes a global shift from infectious diseases to noncommunicable diseases (NCDs), and WHO begins to strengthen its focus on NCDs	First influenza pandemic since 1968 due to new H1N1 influenza virus; WHO works with partners to develop influenza vaccines in record time
2010	2012	2014	2015	2016
WHO issues opinions for raising sufficient resources and removing financial barriers to allow access to essential health services for UHC	For the first time, the WHO Member States set global targets to prevent and control NCDs	The biggest outbreak of Ebola virus disease ever experienced in the world strikes West Africa	2030 Sustainable Development Goals are put forth replace and supersede MDGs	Zika virus among pregnant women represents a Public Health Emergency of International Concern

[a] *TB, malaria, yaws, syphilis, smallpox, and leprosy, etc.*
[b] *The Special Program of Research, Development, and Research Training in Human Reproduction*
[c] *Research and Training in Tropical Diseases; by 2016, five of eight targeted diseases are near elimination*
[d] *The International Conference on Primary Healthcare, in Alma Ata, Kazakhstan*
[e] *Millennium Development Goals*
[f] *Three by five initiative aims to bring treatment to 3 million people living with HIV by 2005*
Adapted from World Health Organization. *The Global Guardian of Public Health*; 2016. http://www.who.int/about/what-we-do/global-guardian-of-public-health.pdf.

Bilateral agencies such as USAID, UKAID, and AusAID are also involved in population, health, and nutrition activities among others. Such programs are done on individual agreements between two governments.

Nongovernmental organizations (NGOs) work to complement governmental agencies. NGOs can be funded by various means—through grants or religious organizations or as partners of larger aid organizations—and they often run small but efficient programs. Examples of NGOs are OxFam, Bill and Melinda Gates Foundation, Clinton Foundation, Wellcome Trust, etc.

Refugee and Disaster Relief Organizations. In contrast to natural disasters, famines and refugee crises tend to develop slowly, often preceded by warning signs of the impending emergency, so that international agencies can coordinate relief efforts with national agencies in a timely fashion. The United Nations agencies are probably the most important of the international relief organizations, but there are several very large NGOs active in refugee and disaster relief. Two of these are the *International Committee of the Red Cross* and *Medecins Sans Frontieres* (commonly called doctors without borders).

Disease Prevention

A major goal of public health practice is prevention.[62] It includes a wide range of activities—known as "interventions"—aimed at reducing risks or threats to health or alters the adverse consequences of the natural history of the disease and health-related events. Such intervention can occur at any time during the natural history of the disease or health-related event.

To prevent disease, we must understand the natural history of the condition (Table 2.4), its distribution in the population, and how to detect early cases. The natural history begins with patients being susceptible to a disease, called the susceptible phase. Susceptible patients can undergo a biological onset of disease. The onset may lie undetected for a varying period called the preclinical phase. When symptoms appear, the patient typically seeks help, perhaps leading to a diagnosis. If therapy begins, this may alter the natural history, turning it into the clinical course of the condition. If there is no treatment, the natural history continues its course, ending in the cure, chronic condition, or decline (death).[63]

There are three identified levels of prevention that are aimed at different stages of disease.[62] Pharmacists have a role to play in all the different stages of prevention (Figs. 2.4 and 2.5).

Primary Prevention: Primary prevention interventions are aimed at individuals or communities in the susceptibility stage directed at reducing the risk of exposure to a risk factor or health determinant in an individual or the population. Immunization is a primary prevention measure. Primordial prevention is a type of primary prevention. Primordial prevention is defined as prevention of risk factors themselves, beginning with a change in social and environmental conditions

TABLE 2.4 An Overview of the Natural History of Diseases and Prevention Strategies to be Applied to Stages of Disease[62]

Stage	Susceptible	Preclinical	Clinical	Recovery, Disability, or Death
Characteristics	Risk factors and disease in the population	Exposure of individuals to risk factor or disease and early pathologic change	Signs and symptoms noted in individuals	Outcomes of clinical disease—death, disability, or recovery
Level of prevention	Primary	Secondary	Tertiary	—
Modes of intervention	Health promotion Specific protection	Screening Early diagnosis	Treatment and rehabilitation	—
Example	Immunization The family lives in adequate, affordable housing	Pap smear Blood pressure measurement in the clinic	Prophylactic medication to prevent myocardial infarction (MI)/prompt administration of thrombolytic therapy following MI	Speech therapy following a cerebrovascular accident

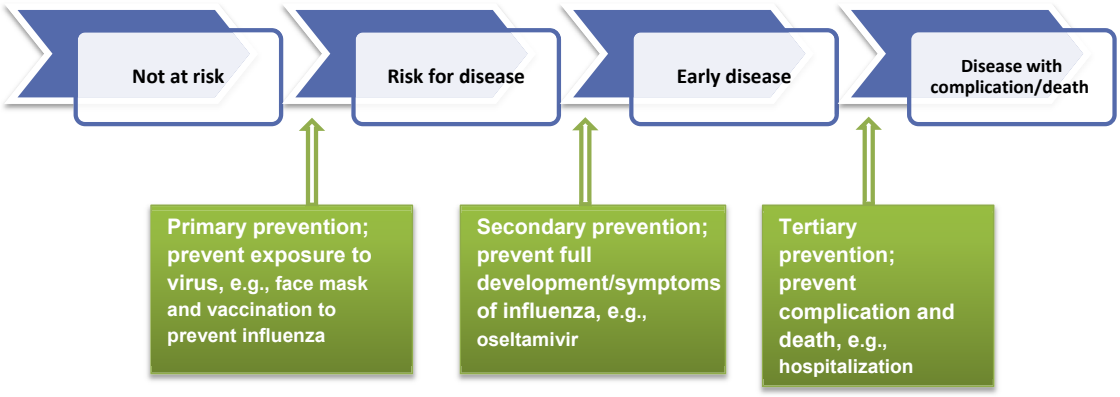


FIGURE 2.4 Levels of disease prevention (e.g., influenza).

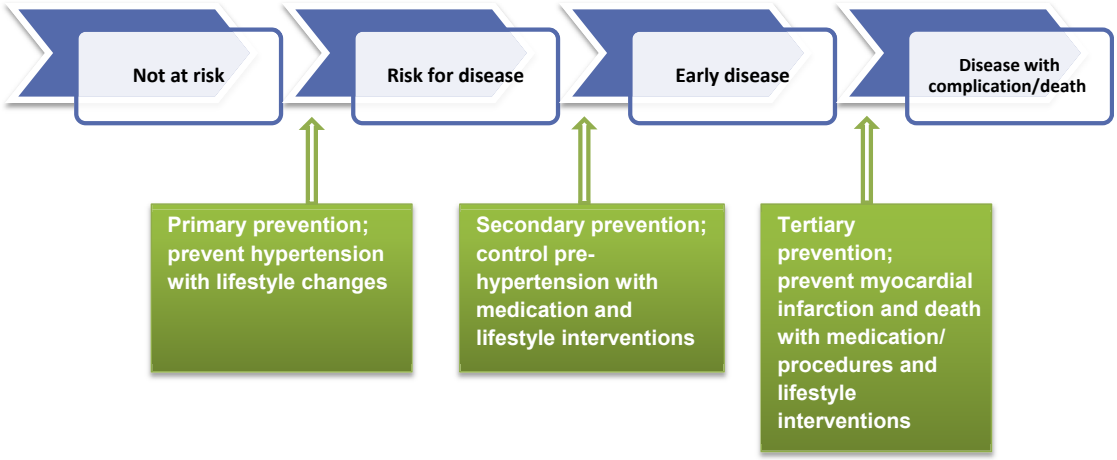


FIGURE 2.5 Levels of disease prevention (e.g., myocardial infarction).

in which these factors are observed to develop and continuing for high-risk children, adolescents, and young adults. Wearing a face mask to avoid exposure to influenza virus is primordial prevention.

Secondary Prevention: Secondary prevention interventions focus on the subclinical stage and the early clinical stage to enable early detection and prompt, effective intervention to correct departures from a state of health. Screening programs such as Pap smears for early diagnosis of cancer of the cervix and mammography for early diagnosis of breast cancer are all examples of secondary prevention. Secondary prevention prevents full development of the disease with symptoms mitigated.

Tertiary Prevention. Tertiary prevention interventions prevent disability or death—although they are used to some extent in any clinical stage. The purpose is to reduce or eliminate suffering, long-term impairments and disabilities, optimize function, assist in adjusting to disease-related limitations in health and function, and extend survival. Examples include prevention of complications and death due to diseases (Figs. 2.4 and 2.5).

Pharmacist's roles in public health continue to evolve. Encouragingly, the American Public Health Association has supported the pharmacist's role in public health as a policy statement.[64]

CONCLUSION

The pharmacy profession developed through identifying, making, and advising how to use drugs. Pharmacists work in a variety of settings such as community pharmacies, healthcare institutions, industries, healthcare agencies, and as consultants. The profession is highly regulated by laws, guidelines, and CoE. The scope of practice has expanded from pharmacy management to additional clinical services to broader public health functions.